Shankhanil Ghosh

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Summary — Generative AI and Full-Stack Engineer with 4+ years of experience building intelligent applications using Python, Java, React, and modern AI frameworks. Hands-on experience with LLM integration, prompt engineering, RAG pipelines, AI-powered automation, and scalable backend systems. Proven ability to develop end-to-end solutions that combine AI capabilities with robust software engineering practices.

Skills

Languages Python, Java, JavaScript, SQL, Bash/Shell Scripting
Backend Frameworks Django, Flask, Spring Boot, REST APIs, Microservices
AI & LLM Technologies Generative AI, LangChain, LangGraph, LlamaIndex, OpenAI API, RAG (Retrieval-Augmented Generation), Prompt Engineering, Vector Databases
Databases PostgreSQL, MySQL, Oracle, SQLite, Redis, ChromaDB
Cloud & AWS AWS EC2, Lambda, S3, CloudWatch, API Gateway, IAM, SNS, SQS, EventBridge
DevOps & Infrastructure Docker, Git, GitHub Actions, CI/CD, Linux, Gunicorn, Nginx
Frontend React.js, HTML5, CSS3, Tailwind CSS, Bootstrap
Tools Postman, DBeaver, VS Code, PyCharm

Experience

Tata Consultancy Services **Nov 2021 – Present**
Application Reliability Engineer

Application Reliability Engineering Portal

- Designed and developed an **Agentic AI-powered incident investigation platform** that dynamically selects and orchestrates diagnostic tools including historical incident retrieval, log analysis, metrics analysis, and dependency analysis to perform root-cause analysis and recommend evidence-backed resolutions.
- Designed and developed an AI-powered **Data Quality Validation Bot** for ServiceNow, improving resolution note quality, data reliability, and enabling the creation of high-quality datasets for ML and LLM training.
- Developed a scalable AI-driven ticket **Auto-Assignment Bot** for ServiceNow and Ultimatix, automating ticket routing, improving OLA compliance, and reducing processing time for 1,000 tickets from 30 minutes to 10 minutes through concurrency, API optimization, and database tuning.
- Implemented AI-based validation and quality scoring mechanisms to detect incomplete, duplicate, and non-compliant ticket data, improving reporting accuracy and operational data quality.
- Designed and implemented a **Gamification** platform for ServiceNow featuring incident management, learning and development, innovation tracking, and performance acceleration modules, increasing employee engagement and improving visibility into operational KPIs.
- Built **DDQube**, a client onboarding and configuration management module that automated onboarding workflows, reduced manual effort, and accelerated adoption across multiple enterprise clients.

AI & Agentic Projects

- **Log Extractor** – Developed an AI-powered agent to parse unstructured system and application logs, identifying critical events, anomalies, and recurring failure patterns to support faster incident analysis.
- **TextToSQL** – Built a natural language–to–SQL generation engine that converts user queries into accurate, executable SQL statements based on predefined database schemas.
- **ProActDoc** – Designed a Generative AI solution to automatically document proactive incident responses, capturing remediation actions, mitigated risks, and operational outcomes.
- **MIMMailer** – Implemented an intelligent email generation agent that produces structured Major Incident (MIM) update communications from incident transcripts and summaries.
- Converted the ARE's ICRM framework from Java to Python and integrated Generative AI features for enhanced automation and data-driven insights.

Internship : Minivet System LLP **Jan 2019 – July 2019**
Junior Software Developer

- Designed and implemented key features, including user authentication, product management, order processing, and payment integration, while optimizing database performance for scalability.
- Collaborated with frontend developers to integrate backend functionality, delivering a secure and seamless online shopping experience.

Projects

FileOps - AI File Organizer GitHub • **July 2026 - Present**

- Built an **AI-powered desktop file management system** using **Python, Streamlit, OpenAI GPT-5, MCP (Model Context Protocol), and ChromaDB** to automatically classify, organize, and search local files.
- Developed a local **MCP server** enabling secure filesystem operations including recursive file discovery, document extraction (PDF, DOCX, TXT), image analysis, file movement, and preview functionality.
- Implemented a **RAG pipeline** with **ChromaDB** to index file metadata and extracted content, enabling natural language search and contextual question answering across thousands of files.
- Integrated multimodal AI using the **OpenAI Vision API** to classify images and documents into hierarchical categories, improving organization accuracy for mixed file types.
- Engineered an end-to-end processing pipeline featuring recursive scanning, metadata extraction, semantic indexing, AI-powered classification, and automated file organization with real-time progress tracking.
- Designed an interactive **Streamlit** interface allowing users to organize files, preview results, perform semantic search, and execute filesystem operations from a single application.

Butterloft butterloft.in • **December 2025 - Present**

- Built and scaled an end-to-end **D2C bakery platform**, managing product catalog, pricing, order lifecycle, and customer engagement.
- Designed and developed a **conversion-focused, responsive UI** to enhance user experience and drive higher purchase rates.
- Integrated secure payment workflows using **Razorpay**, enabling seamless checkout, automated order confirmations, and real-time notifications.
- Implemented logistics and fulfillment workflows including SKU management, shipping label generation, and courier integration.
- Deployed and maintained the platform with custom domain, email infrastructure, and production-grade configurations ensuring high availability and scalability.
- Integrated **Google Analytics** to monitor user behavior, traffic sources, and conversion funnels, supporting **10K+ concurrent users** and enabling data-driven optimization of marketing and product strategies.

DurgaPuja DurgaPuja • **October 2024 - November 2025**

- Developed a full-stack web application for Durga Puja pandal exploration using **React (frontend)** and **Django (backend)**, supporting large-scale user traffic during peak festive periods.
- Built an interactive map-based system to visualize pandal locations and enable users to generate **personalized travel itineraries**.
- Implemented secure and scalable authentication using **OTP, Google, and Twitter login**, improving accessibility and user onboarding.
- Designed a clean, intuitive UI/UX for seamless navigation and efficient trip planning across multiple locations.
- Enabled discovery features for pandals based on location, popularity, and user preferences.
- Integrated **Google Analytics** to track user engagement, peak traffic patterns, and feature usage, supporting **10K+ concurrent users**.
- Deployed on a **DigitalOcean droplet** with domain and email integration via **GoDaddy**, ensuring reliable production performance.

QuickForms Github • **July 2024**

- Designed and developed a full-stack form management platform with real-time response tracking, dynamic form creation, and an intuitive admin dashboard for analytics.
- Implemented data visualization and export features (CSV, Excel), enabling efficient analysis and reporting of user responses.
- Optimized backend queries and data handling for scalable performance across multiple concurrent submissions.

AI-Powered Rental Estimator **June 2024**

- Built a web platform enabling hosts to list properties and receive **AI-driven rental price predictions** based on location, amenities, and seasonal demand.
- Integrated a custom-trained machine learning model to forecast rental income, helping users make data-driven pricing decisions.
- Designed a user-friendly interface for property listing and prediction visualization, improving usability and engagement.

Education

Bachelor's in Technology
University of Engineering and Management Kolkata - 7.73 GPA
High Secondary Education
Army Public School Barrackpore - 81%